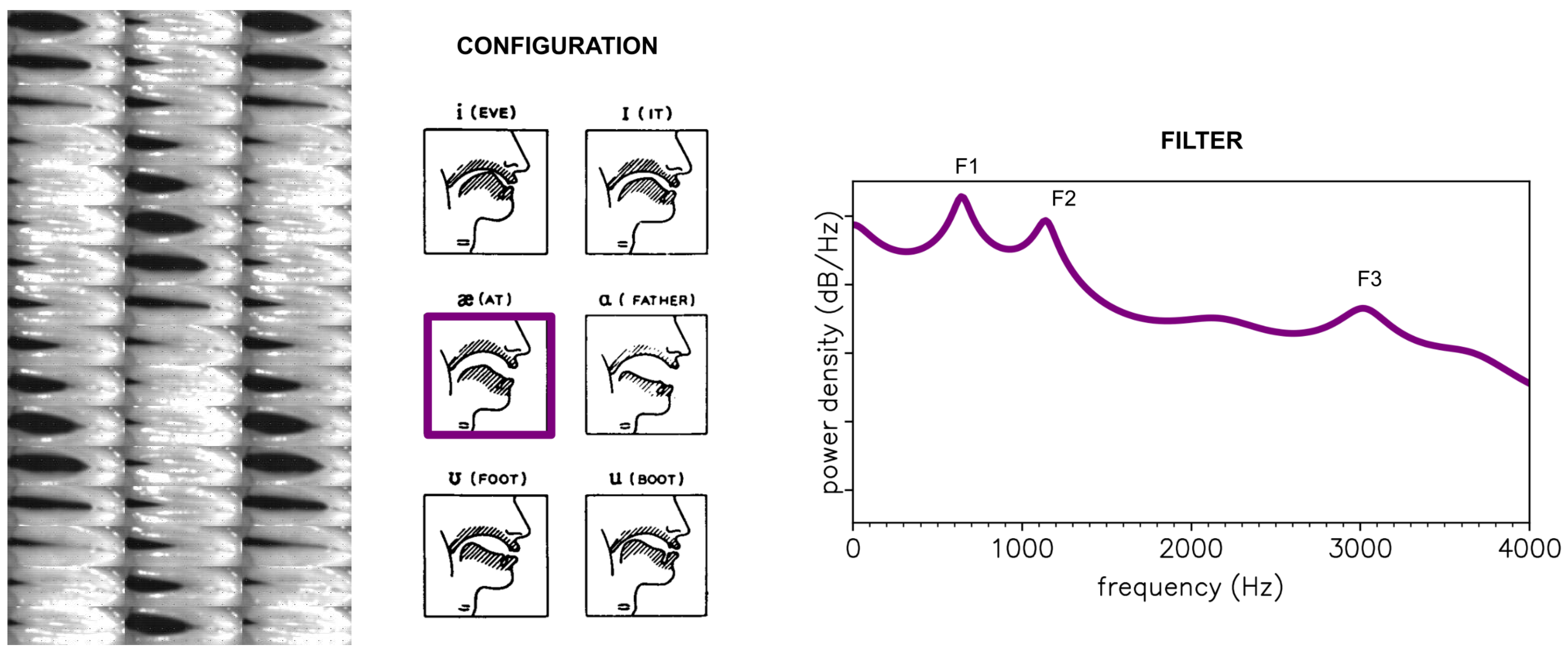
PROBABILISTIC SPEECH DECONVOLUTION IN REAL TIME

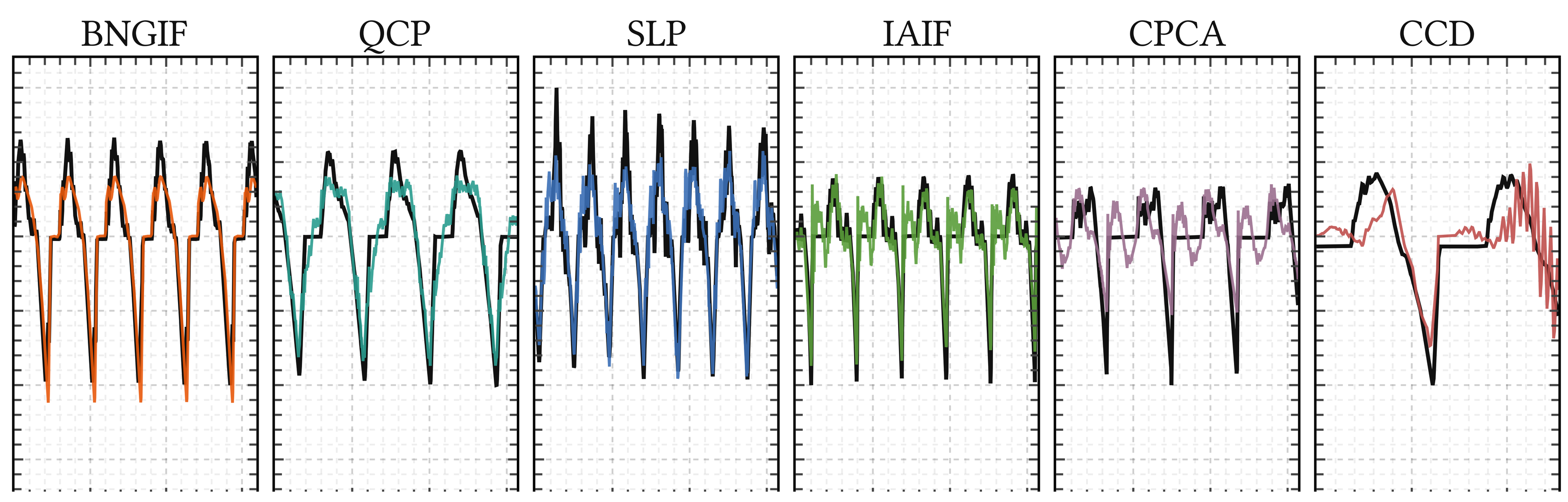
SPEECH = SOURCE × FILTER

Recover the vocal fold airflow and the vocal tract from the microphone alone. Ill-posed, until you say what you know.



Vocal folds (source) and vocal tract (filter, Rabiner & Schafer).

BEATS THE STATE OF THE ART



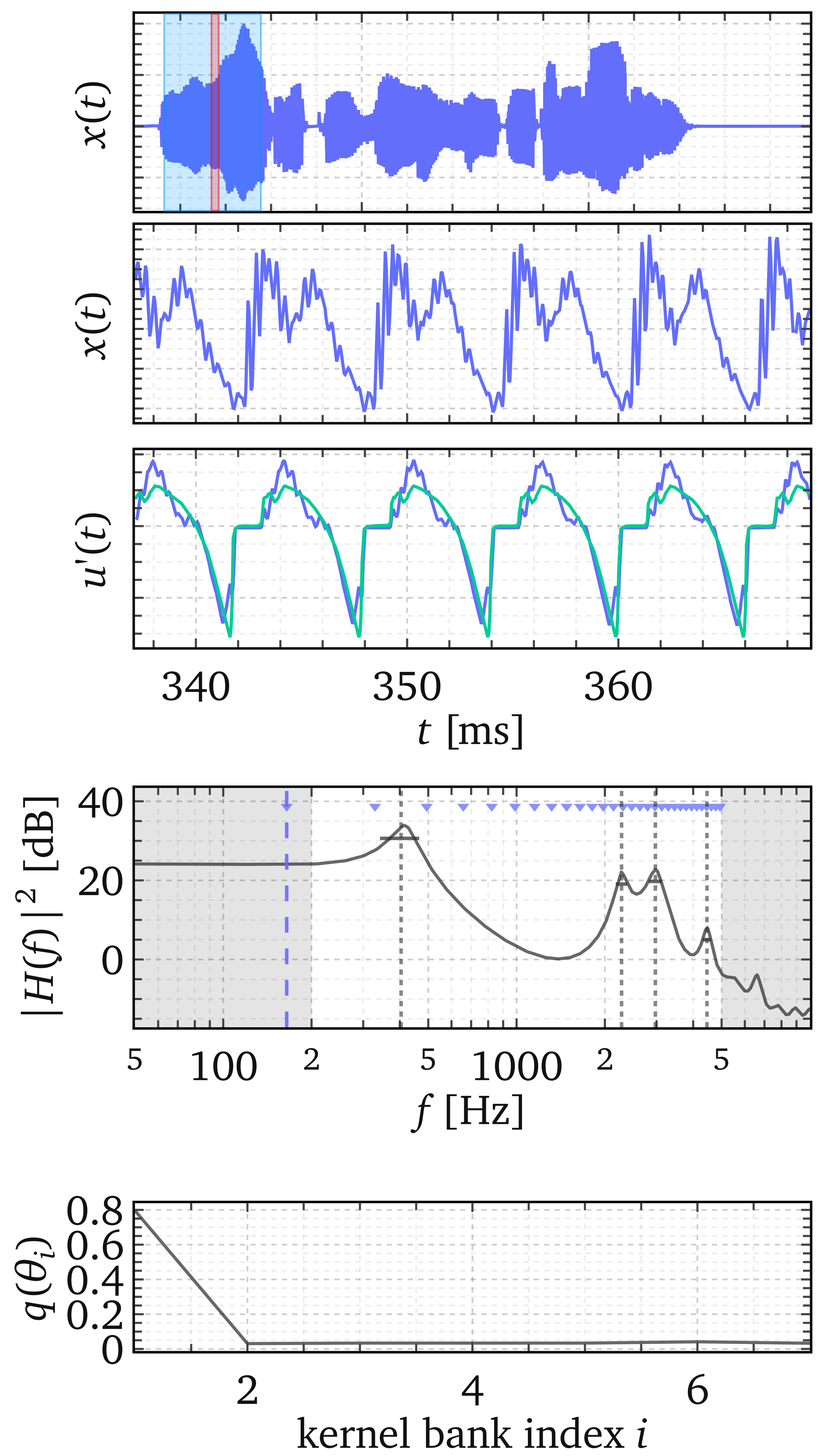
True airflow (black) and each method's estimate, one frame of the EGIFA benchmark.

	BNGIF	QCP	SLP	IAIF	CPCA	CCD
BNGIF	**0.935**	<.001	<.001	<.001	<.001	<.001
QCP	1.000	**0.910**	<.001	<.001	<.001	<.001
SLP	1.000	1.000	**0.889**	<.001	<.001	<.001
IAIF	1.000	1.000	1.000	**0.886**	<.001	<.001
CPCA	1.000	1.000	1.000	1.000	**0.759**	0.004
CCD	1.000	1.000	1.000	1.000	0.996	**0.741**

Paired Wilcoxon over the whole benchmark: BNGIF beats each method, $p < .001$.

ONE MODEL, ALL THE UNCERTAINTY

Posterior over the airflow (band), over the filter, and over which prior shape explains the frame.



5× REAL TIME ON ONE GPU

Variational inference in JAX. A 32 ms frame of 20 kHz speech in 6 ms on an RTX 2080.

WHY IT MATTERS

Hearing implants – pitch and formants with error bars, for sound coding.

Clinics – physiological voice biomarkers instead of black-box features.

Phonetics, forensics – formant tracks 30% more accurate than Praat and KARMA.

VRIJE UNIVERSITEIT BRUSSEL

Marnix Van Soom
AI Lab, Vrije Universiteit Brussel
marnix.van.soom@vub.be

interactive figures